ROGDI Recall Exam Answer Key

1. What disease is the ROGDI case study about?
Answer: Kohlschutter-Tonz syndrome, also called amelocerebrohypohidrotic syndrome.
Topic: Core identity

2. What is the Orphanet ID for the ROGDI/KTS disease case?
Answer: ORPHA:1946.
Topic: Core identity

3. What are the two OMIM identifiers used in the ROGDI audit?
Answer: OMIM phenotype 226750 and OMIM gene 614574.
Topic: Core identity

4. What is the correct gene symbol for the case study?
Answer: ROGDI.
Topic: Core identity

5. What is the correct UniProt accession for human ROGDI?
Answer: Q9GZN7.
Topic: Core identity

6. What is the recommended protein name for UniProt Q9GZN7?
Answer: Protein rogdi homolog.
Topic: Core identity

7. What are three aliases for ROGDI?
Answer: KIAA0267, FLJ22386, and RAV2.
Topic: Core identity

8. What chromosome location is listed for ROGDI?
Answer: 16p12.1.
Topic: Core identity

9. What is the protein length of ROGDI?
Answer: 287 amino acids.
Topic: Core identity

10. What is the approximate molecular weight of ROGDI?
Answer: About 32.3 kDa, or 32,254 Da.
Topic: Core identity

11. What is the approximate ROGDI amino-acid coding region length?

12. Why is source verification the first step before scoring a gene therapy target?
Answer: Because a wrong disease, gene, or protein identity poisons every downstream conclusion: cargo size, mechanism, potency assay, vector fit, and risk assessment.
Topic: Source verification

13. Which source is used to confirm protein identity and sequence facts?
Answer: UniProt. For ROGDI, the key accession is Q9GZN7.
Topic: Source verification

14. Which source is used to anchor rare disease identity?
Answer: Orphanet. For this case, the disease is ORPHA:1946.
Topic: Source verification

15. Which source is used for disease and gene OMIM entries?
Answer: OMIM. For ROGDI/KTS, phenotype 226750 and gene 614574 are used.
Topic: Source verification

16. What live UniProt facts were checked for Q9GZN7?
Answer: Primary accession, protein name, gene name, sequence length, molecular weight, and cross-references such as PDB, AlphaFoldDB, InterPro, and Pfam.
Topic: Source verification

17. What is the inheritance pattern of Kohlschutter-Tonz syndrome?
Answer: Autosomal recessive.
Topic: Disease biology

18. Why does autosomal recessive loss of function support gene addition?
Answer: Because adding a working copy of the gene can, in principle, replace missing or reduced function.
Topic: Disease biology

19. What are the three classic features of KTS described in the audit?
Answer: Amelogenesis imperfecta, infantile or early-onset epilepsy, and intellectual disability or severe neurodevelopmental impairment.
Topic: Disease biology

20. Name three additional clinical features reported in KTS.
Answer: Spasticity, hypohidrosis or heat intolerance, and nephrocalcinosis in some patients.
Topic: Disease biology

21. Why does the audit prioritize CNS rescue?
Answer: Because epilepsy and neurodevelopmental impairment likely dominate morbidity and mortality.
Topic: Disease biology

22. Why might dental disease be hard to reverse in KTS?
Answer: Enamel is formed during a limited developmental window and does not remodel like bone once formed.
Topic: Disease biology

23. How should ROGDI be treated functionally for gene therapy planning?
Answer: As a non-secreted intracellular scaffold or adaptor protein, not as a simple enzyme replacement target.
Topic: Protein function

24. What two PDB structures are linked to human ROGDI?
Answer: 5XQH and 5XQI.
Topic: Protein function

25. What domain or family annotations are listed for ROGDI?
Answer: RAVE2/Rogdi, Rogdi_lz, InterPro IPR028241, and Pfam PF10259.
Topic: Protein function

26. What 2025 biology connection is important for ROGDI?
Answer: ROGDI is connected to Rabconnectin-3-associated V-ATPase biology.
Topic: Protein function

27. Why does the Rabconnectin-3/V-ATPase link matter?
Answer: It gives a plausible mechanism involving acidic organelles, lysosomes, synaptic vesicles, neurons, and enamel/mineralization biology.
Topic: Protein function

28. Is ROGDI secreted?
Answer: No. ROGDI is treated as intracellular and non-secreted.
Topic: Localization

29. Why does non-secreted localization make gene therapy harder?
Answer: Because neighboring corrected cells are unlikely to rescue uncorrected cells by secreting the protein. The affected cells probably need direct transduction.
Topic: Localization

30. What is cell-autonomous rescue?
Answer: A therapeutic effect that requires correction inside the affected cell itself.
Topic: Localization

31. What is cross-correction?
Answer: Rescue of neighboring untransduced cells by a secreted therapeutic protein.
Topic: Localization

32. Why is cross-correction expected to be low for ROGDI?
Answer: Because ROGDI is not known to be secreted.
Topic: Localization

33. Name three reported or annotated ROGDI localization contexts.
Answer: Nuclear envelope, axon/dendrite/perikaryon, presynapse, synaptic vesicle context, or acidic organelle context.
Topic: Localization

34. What is the highest-priority therapeutic tissue for the first ROGDI program?
Answer: CNS neurons.
Topic: Target tissues

35. Which tooth-development cell type matters for KTS dental disease?
Answer: Ameloblasts, the cells responsible for enamel formation.
Topic: Target tissues

36. Why is there a brain/teeth delivery mismatch?
Answer: A CNS-directed route is not expected to robustly correct ameloblasts, and dental-directed delivery will not rescue CNS neurons.
Topic: Target tissues

37. What is AAV?
Answer: Adeno-associated virus, a small non-replicating viral vector commonly used for in vivo gene delivery.
Topic: AAV and cargo

38. What practical AAV packaging limit is used in the audit?
Answer: About 4.7 kb.
Topic: AAV and cargo

39. Why does ROGDI pass the AAV cargo-size gate?
Answer: Its coding sequence is about 861 bp, leaving large room for promoter, polyA, ITRs, and other regulatory elements within the AAV limit.